

CS610 Midterm Paper 2010 of Computer Network - Solved (FQ) with Ref 4

CS610- Computer Network - Midterm Examination - Solved Paper of Spring 2010 - 4

Time: 60 min

Marks= M = 40

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www.vuzs.info/ (<http://www.vuzs.info/>)

<http://groups.google.com/group/vuZs> (<http://groups.google.com/group/vuZs>)

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Question No: 1 (M - 1) . (<http://www.vuzs.info>)

No error detection scheme is perfect because transmission errors can affect the additional information as well as the data.

► **True** (<http://www.vuzs.info/>)

► False

No error detection scheme is perfect because transmission errors can affect the additional information as well as the data. A cyclic redundancy check, which is slightly more difficult to compute, can detect more errors than a Parity or Check sum mechanism.

Question No: 2 (M - 1) . (<http://www.vuzs.info>)

----- Program sends a message to a remote computer and reports whether the computer responds.

► **Ping** (<http://www.vuzs.info/>)

► Traceroute

► ICMP

► Non of the given

Question No: 3 (M - 1) . (<http://www.vuzs.info>)

----- was especially concerned about the lack of high powered computers.

► **ARPA** (<http://www.vuzs.info/>)

► IEEE

► EIA

► Non of the given

The US Department of Defense Advanced Research Projects Agency (ARPA) was especially concerned about the lack of high-powered computers. Many of the ARPA research projects needed access to the latest equipment (<http://www.vuzs.info/>)

Question No: 4 (M - 1) . (<http://www.vuzs.info>)

The term ----- is used to denote the definition of a packet used with a specific type of network.

► Packet

► **Frame** (<http://www.vuzs.info/>)

► Data

► None of the given

To help distinguish between the general idea of packet transmission and the specific definition of packets for a given hardware technology, we use the term frame to denote the definition of a packet used with a specific type of network. Thus one might refer to the maximum amount of data a frame can hold or the order in which bits of a frame are sent across a network.

Question No: 5 (M - 1) . (<http://www.vuzs.info>)

Computer networks are often called ----- because they use packet technology.

- ▶ Ethernet
- ▶ Switch networks
- ▶ (<http://www.vuzs.info/>) Packet networks (<http://www.vuzs.info/>)
- ▶ None of the given

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Question No: 6 (M - 1) . (<http://www.vuzs.info>)

----- have advantages arisen from the size and ease of computation.

- ▶ CRC
- ▶ Parity
- ▶ (<http://www.vuzs.info/>) Checksums (<http://www.vuzs.info/>)
- ▶ None of given

Question No: 7 (M - 1) . (<http://www.vuzs.info>)

Most LANs that employ ring topology use an access mechanism known as-----

- ▶ CSMA/CD
- ▶ CSMA/CA
- ▶ **TOKEN PASSING** (<http://www.vuzs.info/>)
- ▶ None of the given

Most LANs that employ ring technology use an access mechanism known as *token passing*. A token ring operates as a single, shared medium. When a computer wants to send data, it must wait until it obtains the token, when it is in control of the ring

Question No: 8 (M - 1) . (<http://www.vuzs.info>)

IEEE LLC/SNAP header is -----, which is used to specify the type of data.

- ▶ **8 octets** (<http://www.vuzs.info/>)
- ▶ 8 bytes
- ▶ 8 bits
- ▶ None of the given

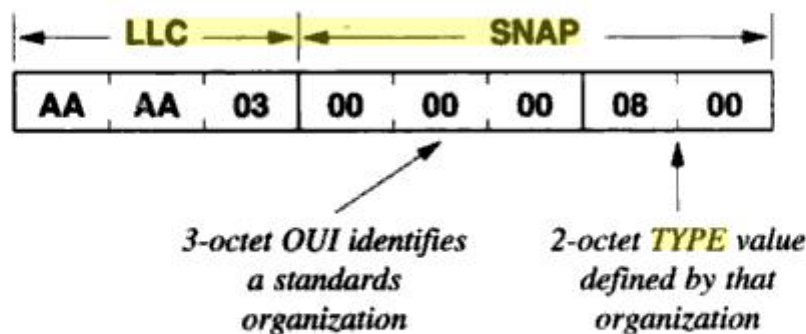


Figure 9.6 An example of the 8-octet **IEEE LLC/SNAP header**, which is **used to specify the type of data**. The SNAP portion specifies an organization and a **type** defined by that organization.

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Question No: 9 (M - 1) . (<http://www.vuzs.info>)

Formally named _____ informally known as the twisted pair Ethernet or TP Ethernet.

- ▶ 10 Base 2
- ▶ 10 Base 5
- ▶ **10 Base T** (<http://www.vuzs.info/>)
- ▶ None of the given

twisted pair ethernet

- Formally called 10 Base –T
- Also twisted pair Ethernet or simply TP Ethernet

Question No: 10 (M - 1) . (<http://www.vuzs.info>)

An interface for twisted pair Ethernet must have an _____ connector , and must generate signals according to the _____ specification.

- ▶ **RJ-45, 10 Base T** (<http://www.vuzs.info/>)
- ▶ RJ-45, 10 Base 5
- ▶ BNC, 10 Base 2
- ▶ BNC, 10 Base T

an interface for twisted pair Ethernet must have an RJ-45 connector, and must generate signals according to the 10BaseT specification. (P# 683 E-books)

Question No: 11 (M - 1) . (<http://www.vuzs.info>)

A bridges function in the _____ layers(s).

- ▶ Physical (MAC)
- ▶ Data link
- ▶ Network
- ▶ (<http://www.vuzs.info>) Physical (MAC) and Data link (<http://www.vuzs.info>)
http://highered.mcgraw-hill.com/sites/0072967722/student_view0/chapter_3_quiz.html (http://highered.mcgraw-hill.com/sites/0072967722/student_view0/chapter_3_quiz.html)

Question No: 12 (M - 1) . (<http://www.vuzs.info>)

A Bridge can _____

- ▶ Filter a frame
- ▶ Forward a frame
- ▶ Extend a LAN
- ▶ **Do all the above** (<http://www.vuzs.info/>)

A bridge is a hardware device also used to connect two LAN segments to extend a LAN. It listens to all traffic and recognizes frame format. It also forwards only correct complete frames and discards the collided and error frames.

The most important task a bridge performs is frame filtering.

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Question No: 13 (M - 1) . (<http://www.vuzs.info>)

A Bridge forwards or filters a frame by comparing the information in its address table to the frame's _____

- ▶ Layer 2 source address
- ▶ Source node's physical address
- ▶ **Layer 2 destination address** (<http://www.vuzs.info/>)
- ▶ Layer 3 destination address

Question No: 14 (M - 1) . (<http://www.vuzs.info>)

_____ computes shortest paths in a graph by using weights on edges as a measure of distance.

- ▶ Greedy algorithm
- ▶ Distance vector algorithm
- ▶ **Dijkstra's algorithm** (<http://www.vuzs.info/>)
- ▶ Non of the given

Dijkstras algorithm computes shortest paths in a graph by using weights on edg as a measure of distance. A path with the fewest number of edges may not be the path with least weight. (P# 173 e-books)

Question No: 15 (M - 1) . (<http://www.vuzs.info>)

_____ is used for audio and video, since these have predefined maximum data rates

- ▶ Constant Bit Rate (CBR) service (<http://www.vuzs.info>)
- ▶ Variable Bit Rate (VBR) service
- ▶ Available Bit Rate (ABR) service
- ▶ None of the given

CONSTANT BIT RATE (CBR):

It is used for audio and video, since these have predefined maximum data rates.

(Omer Chaudhry

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Question No: 16 (M - 1) . (<http://www.vuzs.info>)

Unlike Frame Relay and ATM, SMDS (Switched multi-megabit Data service) offers_____ .

- ▶ **Connectionless service paradigm** (<http://www.vuzs.info/>)
- ▶ Connection oriented service paradigm
- ▶ Both Connectionless and Connection-oriented service paradigm
- ▶ None of the given

Switched Multi megabit Data Service (SMDS) is also a Telco service. It is a connection less service.

FRAME RELAY:

It is used for Telco service for delivering blocks of data. It is connection based

Service ATM uses the concept of connection-oriented networking

Question No: 17 (M - 1) . (<http://www.vuzs.info>)

A network with throughput T and delay D has a total of _____ bit in transit at any time.

- ▶ T / D
- ▶ **T x D** (<http://www.vuzs.info/>)
- ▶ T + D
- ▶ None of the given

A network with throughput T and delay D has a total of T x D bits in transit at any time. (e-books)

Question No: 18 (M - 1) . (<http://www.vuzs.info>)

ATM is designed to work on_____.

- ▶ Twisted Pair
- ▶ Coaxial
- ▶ Radio Frequency
- ▶ **Fiber** (<http://www.vuzs.info/>)

ATM is designed to work on fiber (but can be used with twisted pair). (P# 72)

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Question No: 19 (M - 1) . (<http://www.vuzs.info>)

Computers attached to an ethernet use ----- in which a computer waits for the ether to be idle before transmitting a frame.

(<http://www.vuzs.info/>)

- ▶ **CSMA/CD** (<http://www.vuzs.info/>)
- ▶ CSMA/CA
- ▶ TOKEN PASSING
- ▶ None of the given

All computers attached to the Ethernet use CSMA/CD to co-ordinate their activities. A computer wishing to transmit checks for electrical activity on the cable, informally called a *carrier*. If there is no carrier, the computer can transmit. If a carrier is present, the computer waits for the sender to finish before proceeding

Question No: 20 (M - 1) . (<http://www.vuzs.info>)

FDDI can transmits data at a rate of -----

- ▶ **100 million bits per second** (<http://www.vuzs.info/>)
- ▶ 10 million bits per second

- ▶ 1000 million bits per second
- ▶ None of the given

FDDI: Fiber distributed data interconnect (FDDI) is another ring technology. Its most important features are: It uses fiber optics between stations and transmits data at 100Mbps.

Question No: 21 (M - 2)

What is the difference between the physical and logical topologies?

Question No: 22 (M - 2)

Define Vector-Distance Algorithm.

Question No: 23 (M - 3)

What is the concept of store and forward technology?

Question No: 24 (M - 3)

How can Switched Virtual Network be established?

Question No: 25 (M - 5)

How can a bridge know whether to forward frames?

Question No: 26 (M - 5)

Compare connection oriented and connectionless Service.

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